LABSHEET 05

FORM VALIDATION ARCHITECTURE & CONTROLLED INPUTS IN REACT

Create an functional, responsive simple Login Form module using controlled component inputs inside the React JS library architecture.

Task 5.1: Incorporate dynamic regex criteria verification inside the login controller to trigger interactive error warning badges if parameters violate security patterns.

```
import React, { useState } from "react";
import "./App.css";
export default function App() {
  const [email, setEmail] = useState("");
  const [pass, setPass] = useState("");
  const [step, setStep] = useState(1);
  const [name, setName] = useState("");
  const validEmail = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
  const validPass = /^(?=.*[A-Z])(?=.*\d).{8,}$/;
  let strength = Math.min(pass.length * 10, 100);
  return (
    <div className="box">
      <h2>React Form Validation</h2>
      {step === 1 && (
        <>
          <h3>Login</h3>
          <input
            placeholder="Email"
            value={email}
            onChange={e => setEmail(e.target.value)}
          />
```

```
    {!validEmail.test(email) && email && <p>Invalid Email</p>}
    <input
      type="password"
      placeholder="Password"
      value={pass}
      onChange={e => setPass(e.target.value)}
    />
    {!validPass.test(pass) && pass && (
      <p>Password: 8+ chars, 1 uppercase & 1 number</p>
    )}

    <div className="bar">
      <div style={{ width: strength + "%" }} />
    </div>

    <button onClick={() => setStep(2)}>Next</button>
  </>
)}

{step === 2 && (
  <>
    <h3>Personal Details</h3>
    <input
      placeholder="Name"
      value={name}
      onChange={e => setName(e.target.value)}
    />
    <button onClick={() => setStep(1)}>Back</button>
    <button onClick={() => setStep(3)}>Next</button>
  </>
```

```
      )}

      {step === 3 && (
        <>
          <h3>Confirm</h3>
          <p>Name: {name}</p>
          <p>Email: {email}</p>
          <button onClick={() => alert("Submitted Successfully!")}>
            Submit
          </button>
        </>
      )}
    </div>
  );
}
```

Task 5.2: Structure a password strength checker sub-component that maps text input variations to a dynamic visual progress tracker bar.

```
body {
  font-family: Arial;
  background: #f2f2f2;
}

.box {
  width: 350px;
  margin: 50px auto;
  padding: 25px;
  background: white;
  border-radius: 10px;
}

input {
```

```
  width: 100%;
  padding: 10px;
  margin: 8px 0;
  box-sizing: border-box;
}

button {
  padding: 10px 20px;
  margin: 8px;
  background: #333;
  color: white;
  border: 0;
  border-radius: 5px;
}

p {
  color: red;
  font-size: 13px;
}

.bar {
  height: 8px;
  background: #ddd;
}

.bar div {
  height: 100%;
  background: green;
}
```

Task 5.3: Construct a multi-step user onboarding form flow wizard that encapsulates child components and maintains data states until submittal routines finalize

The onboarding form contains three steps:

Step 1 → Personal Information

Step 2 → Account Information

Step 3 → Confirmation

Onboarding.jsx

```
import React, { useState } from "react";

function Onboarding() {
  const [step, setStep] = useState(1);

  const [formData, setFormData] = useState({
    name: "",
    age: "",
    email: "",
    password: ""
  });
  const handleChange = (e) => {
    const { name, value } = e.target;
  };
  const previousStep = () => {
    if (step > 1) {
      setStep(step - 1);
    }
  };
  return (
    <div className="onboarding">
      {step === 1 && (
        <div className="step-content">
          <h3>Personal Information</h3>
    <input
```

```
      type="text"
      name="name"
      placeholder="Full Name"
      value={formData.name}
      onChange={handleChange}
    />
    <input
      type="number"
      name="age"
      placeholder="Age"
      value={formData.age}
      onChange={handleChange}
    />
    <button onClick={nextStep}>
      Next
    </button>
  </div>
)}

{step === 2 && (
  <div className="step-content">
    <h3>Account Information</h3>
{step === 3 && (
  <div className="step-content">
    <h3>Confirm Details</h3>
    <p>
      <strong>Name:</strong> {formData.name}
    </p>
    <p>
      <strong>Age:</strong> {formData.age}
```

```
          </p>
          <p>
            <strong>Email:</strong> {formData.email}
          </p>
          <button onClick={previousStep}>
            Back
          </button>
          <button onClick={handleSubmit}>
            Submit
          </button>
        </div>
      )}
    </div>
  );
}
```

